

BELGAUM, India

WMO: 431980

Lat: 15.859N

Lon: 74.618E

Elev: 758

StdP: 92.54

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.1	14.3	1.3	4.5	29.5	3.1	5.2	28.6	8.2	26.3	7.5	26.2	0.1	90	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.9	36.5	19.4	35.5	19.5	34.6	19.6	24.3	29.6	23.8	28.7	23.4	28.0	3.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	19.5	26.1	22.6	19.0	25.6	22.2	18.5	25.1	78.0	29.9	75.9	29.0	74.3	28.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.1	7.2	6.2	DB	10.5	38.4	1.0	1.5	9.8	39.5	9.2	40.4	8.6	41.2	7.8	42.3	
			WB	8.0	26.1	1.1	1.1	7.2	26.9	6.6	27.6	6.0	28.2	5.2	29.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	22.1	24.3	26.8	28.4	27.9	24.9	23.4	23.3	23.8	24.3	23.3	22.2
	DBStd	2.46	1.49	1.65	1.71	1.36	1.50	1.69	1.06	1.00	1.16	1.11	1.44	1.48
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5311	376	399	522	553	556	446	414	412	413	444	398	377
	CDD18.3	2269	118	166	263	303	298	196	156	153	163	186	148	119
	CDH23.3	17628	1097	1749	2939	3242	2879	1024	398	417	696	1176	1073	936
	CDH26.7	7327	329	816	1622	1813	1473	283	32	36	128	324	262	208
Wind	WSAvg	2.5	1.6	1.7	1.9	2.2	3.1	3.8	3.9	3.3	2.4	1.6	1.9	2.0
Precipitation	PrecAvg	1035	2	2	12	43	90	158	279	185	118	104	35	6
	PrecMax	1634	47	29	124	122	397	356	720	613	295	274	179	47
	PrecMin	559	0	0	0	0	4	20	63	57	11	3	0	0
	PrecStd	294	8	6	22	30	77	63	144	113	69	64	41	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	34.7	36.7	37.8	37.4	33.6	28.9	29.1	30.8	31.6	30.8	30.9
		MCWB	17.3	17.4	18.2	19.7	19.9	22.8	22.9	23.0	22.4	19.9	19.0	18.6
	2%	DB	30.6	33.5	35.6	36.8	36.1	31.2	27.5	27.7	29.3	30.3	29.7	29.5
		MCWB	17.2	17.3	18.1	19.5	20.5	23.0	22.6	22.6	22.4	20.2	19.1	18.0
	5%	DB	29.6	32.5	34.8	35.8	34.7	29.6	26.5	26.7	28.2	29.5	29.0	28.7
		MCWB	16.9	17.4	18.1	19.7	21.3	23.0	22.4	22.4	22.4	20.4	19.1	17.7
	10%	DB	28.7	31.4	33.9	34.8	33.5	28.1	25.5	25.6	27.1	28.6	28.3	27.9
		MCWB	16.8	17.3	18.3	20.0	21.8	22.7	22.3	22.2	22.3	20.8	19.1	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.7	23.5	24.7	25.1	24.6	23.6	23.7	24.2	24.0	22.9	21.9
		MCDB	26.6	28.6	29.0	30.7	31.3	29.3	26.9	27.3	28.2	28.1	27.1	27.5
	2%	WB	19.8	20.6	22.6	23.8	24.4	23.8	23.1	23.1	23.5	23.2	22.2	21.0
		MCDB	26.2	27.8	27.8	29.4	30.6	28.1	26.1	26.3	27.2	27.1	26.4	26.0
	5%	WB	19.0	19.7	21.8	23.2	23.8	23.4	22.7	22.7	23.0	22.8	21.7	20.2
		MCDB	25.4	27.2	27.1	28.5	29.8	27.5	25.4	25.6	26.5	26.4	25.7	25.2
	10%	WB	18.2	19.0	21.1	22.7	23.3	23.0	22.4	22.3	22.6	22.4	21.2	19.5
		MCDB	25.2	26.7	26.7	27.9	28.9	26.8	24.8	24.9	25.7	25.7	25.1	24.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.7	14.8	13.9	12.1	6.4	4.4	4.9	6.9	8.9	10.8	12.4
		MCDBR	14.9	15.9	15.4	15.0	14.0	9.3	6.3	6.8	9.0	10.5	11.7	13.3
		MCWBR	4.9	5.2	5.3	5.0	4.0	2.7	2.4	2.7	3.2	3.5	4.1	4.7
	5% WB	MCDBR	12.3	14.2	14.2	13.6	12.0	7.4	5.0	5.6	7.2	8.1	8.6	10.3
MCWBR		5.0	5.4	5.1	4.6	3.9	2.9	2.4	2.7	3.1	3.2	3.4	4.3	
Clear-Sky Solar Irradiance	taub		0.404	0.418	0.496	0.562	0.535	0.516	0.536	0.504	0.467	0.517	0.459	0.418
	taud		2.239	2.224	2.016	1.890	1.981	2.078	2.011	2.112	2.232	2.025	2.148	2.218
	Ebn at Noon		882	887	826	771	780	787	774	807	840	785	820	855
	Edh at Noon		134	142	180	204	184	165	177	161	143	171	146	134
All-Sky Solar Radiation	RadAvg		5.28	6.08	6.58	6.82	6.49	4.37	3.68	3.88	4.56	4.96	4.97	4.99
	RadStd		0.21	0.19	0.26	0.23	0.36	0.40	0.40	0.26	0.43	0.27	0.31	0.18

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.15	N/A	+0.86	+0.37	+0.36	+0.34	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page